



ACTIVITY COMPLETE

Great work, Abhiram Chittampally!

930

Total points scored

Review your answers

19 of 20 correct



Q1

What is the primary purpose of a fact table in a dimensional model?

Your answer: B. Store measurable business events along with foreign keys to dimensions

**Correct answer: B. Store measurable business events along with foreign keys to dimensions**

Q2

What does the abbreviation SCD stand for in data warehousing?

Your answer: C. Slowly Changing Dimension

**Correct answer: C. Slowly Changing Dimension**

Q3

Which schema normalizes dimension tables into hierarchical lookup tables to reduce redundancy?

Your answer: B. Snowflake Schema

**Correct answer: B. Snowflake Schema**

Q4

In an SCD Type 2 dimension, which column typically identifies the latest active record for a given business key?

Your answer: D. IS_CURRENT

**Correct answer: D. IS_CURRENT**

Q5

Why do Kimball dimension tables use a surrogate key instead of the source system's business key?

Your answer: C. To provide a warehouse-generated unique key independent of source system changes and to support SCD versioning

**Correct answer: C. To provide a warehouse-generated unique key independent of source system changes and to support SCD versioning**

Q6 In **FACT_ADMISSION** for the hospital data warehouse, which of the following is an additive measure?

Your answer: C. LENGTH_OF_STAY



Correct answer: C. LENGTH_OF_STAY

Q7 The bus matrix above lists dimensions and the two fact tables of the hospital data warehouse. Which dimension is NOT conformed between **FACT_ADMISSION** and **FACT_BILLING**?



Verifying image...

Checking the student-facing preview.

Bus Matrix — Hospital Data Warehouse

Dimension	FACT_ADMISSION	FACT_BILLING
DIM_PATIENT	✓	✓
DIM_DOCTOR	✓	✓
DIM_HOSPITAL	✓	✓
DIM_DEPARTMENT	✓	✓
DIM_DATE	✓	✓
DIM_TREATMENT	—	✓

✓ Optimized image preview

Click Save to keep this image with the activity.

Your answer: C. DIM_TREATMENT



Correct answer: C. DIM_TREATMENT

Q8 The SQL statement below is executed against the customer dimension. Which SCD strategy does it implement?

SQL

```
UPDATE DIM_CUSTOMER
SET   CITY    = 'Bengaluru',
      STATE   = 'Karnataka',
      MEMBERSHIP = 'Gold'
WHERE CUSTOMER_ID = 101;
```

Your answer: A. SCD Type 1 — overwrite the existing row, history is lost



Correct answer: A. SCD Type 1 — overwrite the existing row, history is lost

Q9

In the retail Snowflake Schema for Project 6, DIM_BRANCH is connected directly (via a foreign key) to which of the following lookup tables?

Your answer: C. DIM_CITY



Correct answer: C. DIM_CITY

Q10

The customer dimension starts with 5 rows. Three customers experience attribute changes and SCD Type 2 is applied. How many rows will the Type 2 dimension contain after the updates?

Your answer: C. 8 rows



Correct answer: C. 8 rows

Q11

Which statement best defines the grain of FACT_ADMISSION in the hospital data warehouse?

Your answer: D. One row per patient admission to one hospital under one doctor and department on one admission date



Correct answer: D. One row per patient admission to one hospital under one doctor and department on one admission date

Q12

Which of the following correctly describes SCD Type 3?

Your answer: B. Maintains only the current and previous values in extra columns without creating a new row



Correct answer: B. Maintains only the current and previous values in extra columns without creating a new row

Q13

Which Kimball data warehousing concept is being demonstrated by the query below?

SQL

```
SELECT h.hospital_name,
       COUNT(a.admission_key) AS total_admissions,
       SUM(b.net_amount) AS total_revenue
FROM   dim_hospital h
JOIN   fact_admission a ON h.hospital_key = a.hospital_key
JOIN   fact_billing b ON h.hospital_key = b.hospital_key
GROUP BY h.hospital_name;
```

Your answer: C. Drill-across analysis using a conformed dimension



Correct answer: C. Drill-across analysis using a conformed dimension

Q14

In the retail Snowflake Schema, the Product Hierarchy is arranged (from the fact-connected dimension outward to the highest lookup) as:

Your answer: B. DIM_CATEGORY → DIM_PRODUCT → DIM_BRAND



Correct answer: A. DIM_PRODUCT → DIM_BRAND → DIM_CATEGORY

Q15

A retail chain must track a customer's complete membership history AND also be able to fetch the customer's current and immediately previous membership from a single row without joining historical versions. Which SCD strategy best satisfies both requirements?

Your answer: D. SCD Type 6



Correct answer: D. SCD Type 6

Q16

Customer 101's SCD Type 2 rows are: (Hyderabad, Silver, EFFECTIVE_DATE 2026-01-01, EXPIRY_DATE 2026-03-31, IS_CURRENT FALSE) and (Bengaluru, Gold, EFFECTIVE_DATE 2026-04-01, EXPIRY_DATE 9999-12-31, IS_CURRENT TRUE). What does the query below return?

SQL

```
SELECT customer_id, city, membership,  
       effective_date, expiry_date  
FROM   dim_customer_type2  
WHERE  customer_id = 101  
       AND DATE '2026-03-15' BETWEEN effective_date AND expiry_date;
```

Your answer: B. One row: Hyderabad, Silver



Correct answer: B. One row: Hyderabad, Silver

Q17

Which of the following correctly compares Star Schema and Snowflake Schema on storage and query performance?

Your answer: A. Star uses more storage due to redundant denormalized dimensions but is generally faster because fewer joins are required; Snowflake uses less storage but is slightly slower due to more joins



Correct answer: A. Star uses more storage due to redundant denormalized dimensions but is generally faster because fewer joins are required; Snowflake uses less storage but is slightly slower due to more joins

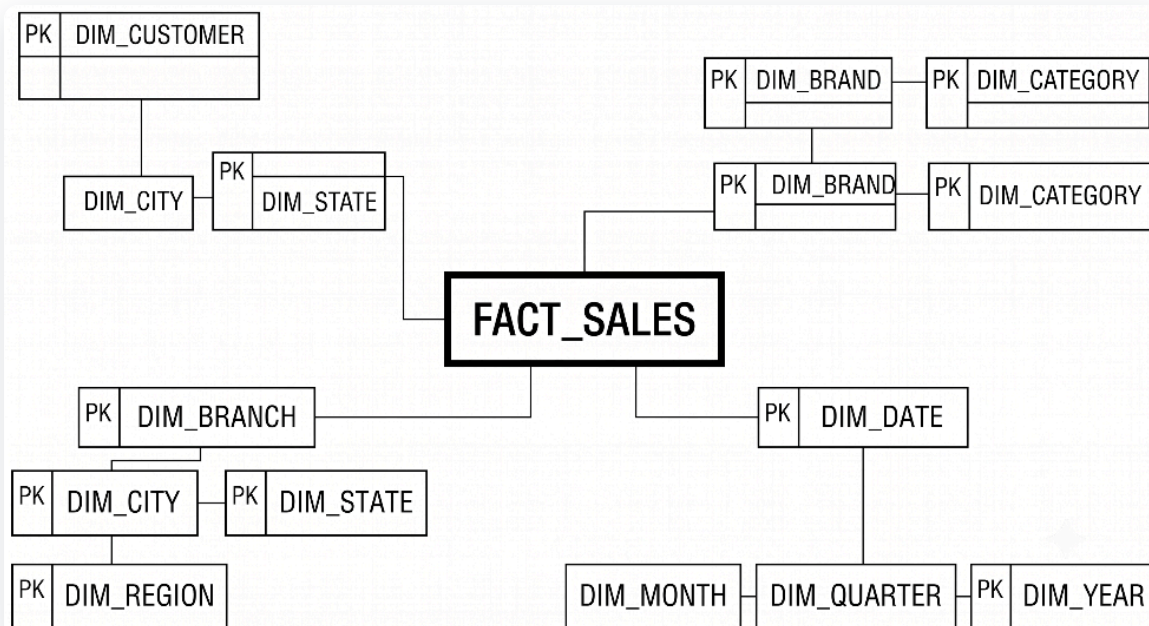
Q18

The schema diagram shown above represents which type of data warehouse schema?



Verifying image...

Checking the student-facing preview.



✓ Optimized image preview

Click Save to keep this image with the activity.

Your answer: B. Snowflake Schema



Correct answer: B. Snowflake Schema

Q19

Hospital management asks: 'How many admissions AND how much revenue did KMIT Hospital generate in Q1 2026?' Using Kimball conformed dimensions, what is the minimum required set of fact tables and shared dimensions to answer this in a single drill-across report?

Your answer: C. Both FACT_ADMISSION and FACT_BILLING, each joined to the same conformed DIM_HOSPITAL and DIM_DATE



Correct answer: C. Both FACT_ADMISSION and FACT_BILLING, each joined to the same conformed DIM_HOSPITAL and DIM_DATE

Q20

The customer dimension is loaded initially with 5 customers. Later, 3 of those customers each experience one membership change and SCD Type 6 is applied. What will the Type 6 dimension contain after all changes are processed?

Your answer: B. 8 rows in total — 5 marked as current and 3 historical rows



Correct answer: B. 8 rows in total — 5 marked as current and 3 historical rows